# Estimating Potato Nitrogen Content Using Active Transfer Learning Based on UAV Hyperspectral Imagery

Hang Yin, Haibo Yang, Fei Li, Yuncai Hu, and Kang Yu

***Abstract*—Significant progress has been made in estimating plant nitrogen content (PNC) using remote sensing data, driven by advancements in machine learning models. However, external factors such as interannual variability, soil heterogeneity, and fertilization management often influence model stability across different growing seasons. Transfer learning has gained attention as an effective strategy to enhance model generalization, yet the domain adaptability of transferred samples remains a pressing challenge that warrants further investigation. Therefore, this study integrated active learning with transfer learning to identify the most informative transferable samples, thereby enhancing model generalization across heterogeneous growth environments. Spectral data of potato plants were collected using uncrewed aerial vehicles (UAVs)-based hyperspectral imaging at three locations from 2018 to 2021. The transferability of an unsupervised transfer method [transfer component analysis (TCAs)] and two sample-based transfer methods (random sample update and active learning sample update) combined with partial least squares regression (PLSR) was compared across different years. The results showed that both sample-based transfer methods outperformed the unsupervised transfer method across different years. The random sample update required 35% of the target dataset to reach moderate accuracy ($R^2$ = 0.40–0.70), whereas the active learning sample update attained superior performance ($R^2$ = 0.48–0.75) using only 5–6 high-residual samples. This approach effectively minimized latent space misalignment within the PLSR model, thereby enhancing the model's transferability to unseen datasets. Furthermore, the active learning sample update method identified key nitrogen-sensitive spectral bands, primarily within the 520–570 nm and 68–730 nm ranges. Thus, our study suggests that the active learning sample update could significantly advance PNC estimation and support scalable PNC monitoring using future hyperspectral satellite missions.**

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***Index Terms*—Active learning, hyperspectral, nitrogen, potato, transfer learning.**

## I. Introduction

POTATO (*Solanum tuberosum L.*) ranks the fourth most important staple crop worldwide and plays a vital role in global food security and the agricultural economy [1], [2]. Plant nitrogen content (PNC) not only directly influences potato tuber yield and quality but also is closely linked to *N* use efficiency and agroecosystem sustainability [3], [4]. Accurate estimation of potato PNC is therefore essential for enabling precise *N* management, increasing yield, and mitigating *N* losses. In recent years, remote sensing technology has been widely applied in crop PNC estimation due to its rapid, noninvasive, and large-scale monitoring capabilities [5], [6], [7]. However, across different years and experimental sites, crops are subjected to substantial variability in climatic conditions (e.g., temperature and precipitation), soil characteristics, and nutrient and water management regimes. These factors jointly govern nitrogen uptake dynamics in crops, making it challenging for PNC estimation models trained under specific conditions to retain estimation reliability in different environmental settings. Conventional approaches typically require extensive data collection and model retraining for each new context, which is labor-intensive, resource-demanding, and restricts the model's scalability. Consequently, enhancing the transferability of PNC estimation models across diverse growing conditions and management practices has become a critical challenge in the pursuit of precision agriculture and sustainable nutrient management.

Spectral indices (SIs) are among the most widely used methods for estimating PNC [8], [9]. SI-based approaches are particularly suitable for multispectral data and can yield reliable estimates; however, uncertainties and saturation effects in the relationship between SIs and PNC often undermine their robustness, limiting their broader application. In contrast, machine learning algorithms, such as partial least squares regression (PLSR), have been widely employed for crop PNC estimation [10]. PLSR leverages the higher spectral dimensionality of hyperspectral data, enabling comprehensive analysis by exploiting spectral information that SIs-based methods cannot easily achieve. Numerous studies have demonstrated that PLSR models can accurately estimate crop yield [1], chlorophyll content [11], and PNC [12]. Although PLSR performs well for dataset-specific scenarios, its generalizability diminishes when applied to new datasets with different feature spaces and distributions under varying growth conditions. For

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instance, Liu et al. [13] trained the PLSR model using full-spectrum data from 2019 to estimate potato parameters; when applied to the 2018 dataset, the model exhibited limited estimation performance ($R^2$ = 0.49), likely due to environmental variability. Furthermore, differences among spectral sensors in spectral resolution, illumination conditions, and viewing geometry may also impair the model's generalizability to new datasets.

In recent years, transfer learning has exhibited distinctive advantages in agricultural remote sensing and crop growth modeling [14], [15]. Unlike conventional machine learning methods that typically rely on large quantities of labeled data collected under specific conditions, transfer learning enables the reuse of features learned from source datasets in target datasets through knowledge transfer, thereby mitigating the limitations posed by sample scarcity and domain shift. In PNC estimation, models trained on data from a single year or specific conditions often exhibit reduced estimation consistency across interannual scales, largely due to the influence of climate variability, soil heterogeneity, and fertilization management practices. Transfer learning enhances model robustness and generalization by facilitating the sharing and adaptation of feature representations across diverse environmental settings, effectively addressing the uncertainty and instability inherent in interannual estimation. Currently, two widely used transfer learning strategies include unsupervised transfer learning [transfer component analysis (TCA)] and sample-based transfer learning approaches (random sample updating) [16]. TCA is a representative feature space transfer method that projects the source and target datasets into a common low-dimensional subspace, thereby minimizing distribution discrepancies between them. Previous studies have shown that integrating TCA with support vector regression (SVR) can enhance the transferability of leaf nitrogen content estimation across different species [16]. Nevertheless, compromised. TCA's effectiveness may be compromised when the distribution gap between the source and target datasets is significant. In contrast, the random sample update method can effectively mitigate the challenge of significant distribution discrepancies between datasets. As an instance transfer technique, random sample update incrementally improves model generalizability by updating the existing model with randomly selected samples from the target dataset [17]. Recent studies have demonstrated that this method outperforms TCA and other transfer approaches, significantly improving model transferability across different species [16], environments [17], and locations [18]. Despite its promise for enhancing crop PNC estimation, a critical challenge remains: efficiently identifying samples with positive transfer value. Inappropriate sample selection may result in nonrepresentative samples that differ significantly from the target dataset, thereby reducing model performance and potentially causing adverse transfer effects.

By strategically assembling the most informative training set, active learning has emerged as a promising approach to addressing this challenge. Within an active learning framework, a model is trained on a small and initially suboptimal training set, then iteratively selects samples from a large pool of unlabeled data based on a predefined acquisition function. These samples are annotated by external experts and incorporated into the training set. This process is repeated until the labeling budget is exhausted or the target accuracy is achieved. Unlike most passive learning models, active learning does not select training samples at random but is designed to identify those most conducive to improving model performance. In recent years, active learning methods have been successfully applied in fields such as image classification [19], soil and crop mapping [20], and crop parameter estimation [21]. In crop PNC estimation, active learning has primarily been used for selecting simulated datasets within hybrid inversion models. Verrelst et al. [22] and De Clerck et al. [23] proposed a hybrid inversion framework for PNC estimation, in which the model was trained on a large-scale dataset of spectra and biophysical parameters generated by the PROSAIL model. Consequently, they optimized via active learning to select the simulated dataset with optimal generalizability. However, simulated data differ markedly from in situ measurements in noise characteristics, sensor responses, and environmental complexity. As a result, even when optimized through active learning, simulated datasets may still limit model performance in real-world applications. By contrast, in situ samples more faithfully represent the characteristics of the target environment, yet their collection is often constrained by labor, time, and cost. Incorporating active learning into this context enables the efficient selection of the most informative samples from limited in situ data, thereby substantially enhancing model generalizability and estimation accuracy. To the best of our knowledge, few studies have leveraged active learning to select the most informative transfer samples, improving model transferability and estimation accuracy under limited-data conditions.

Machine learning models often perform well on specific datasets but exhibit limited transferability when applied to new datasets. While existing transfer learning approaches have made notable progress, they remain inadequate in identifying the most informative samples. To overcome these limitations, therefore, this study proposed a method that integrates active learning with a random sample update strategy. By using active learning to dynamically optimize the selection of source dataset samples, the method seeks to enhance model transferability across different locations and years. The specific objectives of this study were: 1) to evaluate the effectiveness of multiple transfer learning approaches in improving machine learning model performance; 2) to propose a method based on active transfer learning; and 3) to construct a generalized multiyear potato PNC estimation model based on active transfer learning and evaluate its applicability across sensors with varying spatial resolutions.

## II. Materials and Methods

The flowchart of this study is shown in Fig. 1. This study mainly included the acquisition of hyperspectral images and PNC data, the performance evaluation of different transfer learning models across years, and the application of transfer learning models across sensors.

### A. Study Area and Experimental Design

In 2018, 2020, and 2021, potato trials were conducted in Zhuozi and Wuchuan Counties in Inner Mongolia, China (Fig. 2). The potato growing season in this region typically

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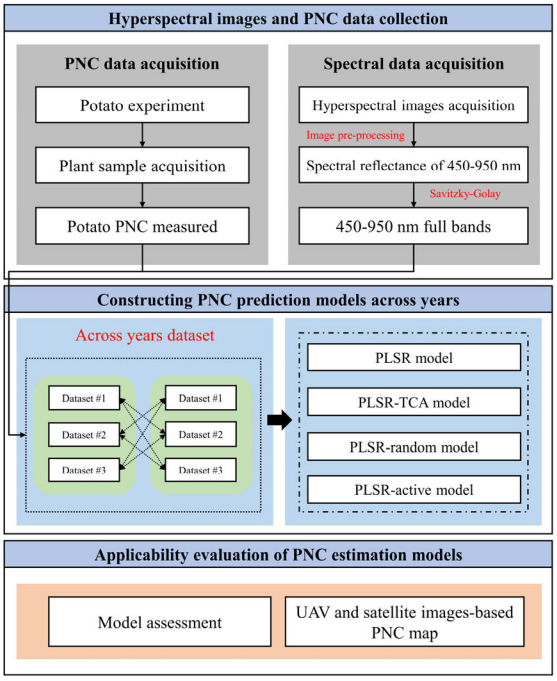



Fig. 1. Workflow of this work.

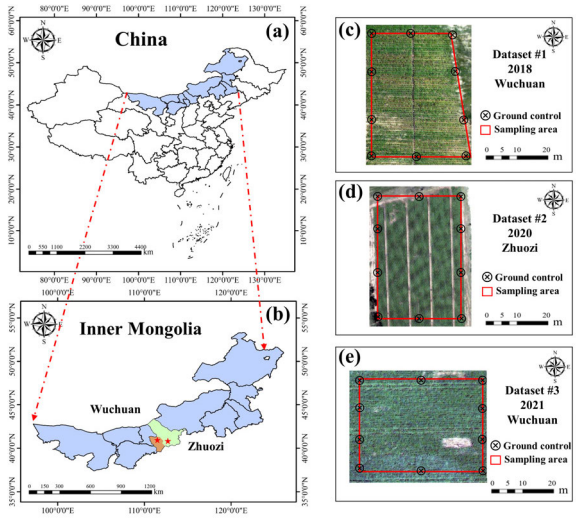



Fig. 2. (a) and (b) Location of the study area in Wuchuan and Zhuozi and (c)–(e) experimental layout in 2018, 2020, and 2021.

spans from June to September. Different years, *N* rates, and cultivars were investigated (Table I). Each experimental treatment consisted of four replicates, and each plot area was 120 $m^2$. The application of fertilizers other than *N* followed the conventional practices of local farmers.

### B. Hyperspectral Image

The uncrewed aerial vehicle (UAVs) imaging system used in this study included a multirotor DJI Matrice 600 Pro (DJI, Shenzhen, China) and a hyperspectral imaging spectrometer (S185, Cubert GmbH, Ulm, Baden Württemberg, Germany) (Fig. 3). The hyperspectral imaging spectrometer had a spectral range of 450–950 nm, with 125 spectral bands and a spectral interval of 4 nm. Hyperspectral images were acquired simultaneously with the plant samples during the same period, between 10:00 A.M. and 2:00 P.M. on the same day. The overlap rate of the hyperspectral images was 80%. The UAV flew at an altitude of 30 m, resulting in a spatial resolution of 7.8 cm. The UAV's flight speed is 6 m/s. Before the UAV flight operation, the spectrometers were radiometrically calibrated using a standard reflectance whiteboard. The hyperspectral images were stitched and processed using the Cube-Pilot and Agisoft PhotoScan software. The processed hyperspectral images were imported into ENVI 5.6 software. The average spectral reflectance of the potato plants in each plot was extracted using the region of interest tool. The Savitzky–Golay filter, using a quadratic polynomial and a window size of 21, was employed to smooth the extracted hyperspectral reflectance data, thereby improving spectrum smoothness and reducing noise [13]. The hyperspectral reflectance of the potato was shown in Fig. 4.

### C. PNC Sample Collection

Plant samples were collected during the critical growth stages of the potato. The specific sampling dates for the potato are shown in Table I. Sampling was conducted on two consecutive rows of potato plants, each 1 m apart. The aboveground plant samples were chopped and mixed, and subsamples of 400–600 g were taken. All subsamples were dried at 70 °C until a constant weight, weighed, ground, and then used for chemical analysis. PNC was determined using the Kjeldahl-N analysis method. The PNC values of potato are shown in Fig. 4.

### D. Sentinel-2A Image Acquisition and Preprocessing

Satellite-based remote sensing has been widely applied for monitoring large-scale PNC due to its broad spatial coverage and revisit capability [24], [25]. In this study, we evaluated the transferability of PNC inversion models from UAV-scale hyperspectral imagery to satellite-scale observations. Sentinel-2A imagery was acquired and preprocessed following the procedures described in our previous study [26]. The UAV-based hyperspectral sensor acquired reflectance data in the wavelength range of 450–950 nm with a spectral sampling interval of 4 nm. The original spectral resolution of the UAV hyperspectral data before resampling was approximately 4 nm, whereas the Sentinel-2A multispectral imagery has discrete bands with bandwidths ranging from approximately 15 to 20 nm in the visible and near-infrared (VNIR) region. To ensure spectral consistency between UAV and satellite data, the UAV hyperspectral reflectance spectra were resampled to match the spectral configuration of Sentinel-2A. Specifically, band-averaging was performed by averaging the reflectance values of all UAV bands that fell within the spectral response range of each Sentinel-2A VNIR band. Sentinel-2A bands with central wavelengths beyond 950 nm were excluded from further analysis due to the limited spectral coverage of the UAV sensor. This resampling process ensured that the UAV and satellite datasets were spectrally comparable for subsequent modeling [27].

### E. Partial Least Squares Regression

We chose PLSR as the baseline for model transferability comparison due to its strong interpretability, which enables

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TABLE I
SUMMARY OF THE FIELD EXPERIMENT DESIGN IN DIFFERENT LOCATIONS

| | Site | Cultivar | N fertilizer application rates | Sampling date |
|---|---|---|---|---|
| Dataset #1 | Wuchuan | Jizhangshu 12 | N rate (kg ha$^{-1}$): 0, 81, 162, 243, 324 | 2018/07/25<br>2018/08/10<br>2018/08/18 |
| Dataset #2 | Zhuozi | Innovator | N rate (kg ha$^{-1}$): 198, 204, 264, 313, 373 | 2020/07/20<br>2020/08/04<br>2020/08/20 |
| Dataset #3 | Wuchuan | Jizhangshu 12 | N rate (kg ha$^{-1}$): 0, 60, 80, 151, 238, 325 | 2021/07/23<br>2021/08/12<br>2021/08/29 |

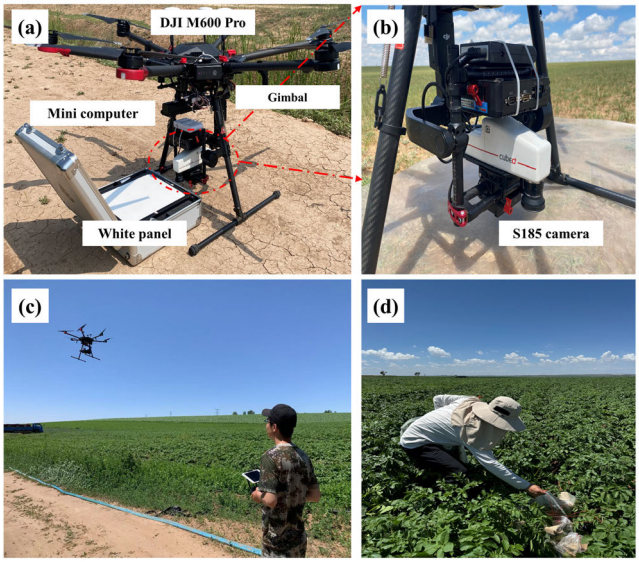



Fig. 3. Data collection scenario. (a) UAV hyperspectral system consists of a UAV. (b) Hyperspectral camera. (c) and (d) Ground measurement data collection is performed on the same day as the UAV flight.

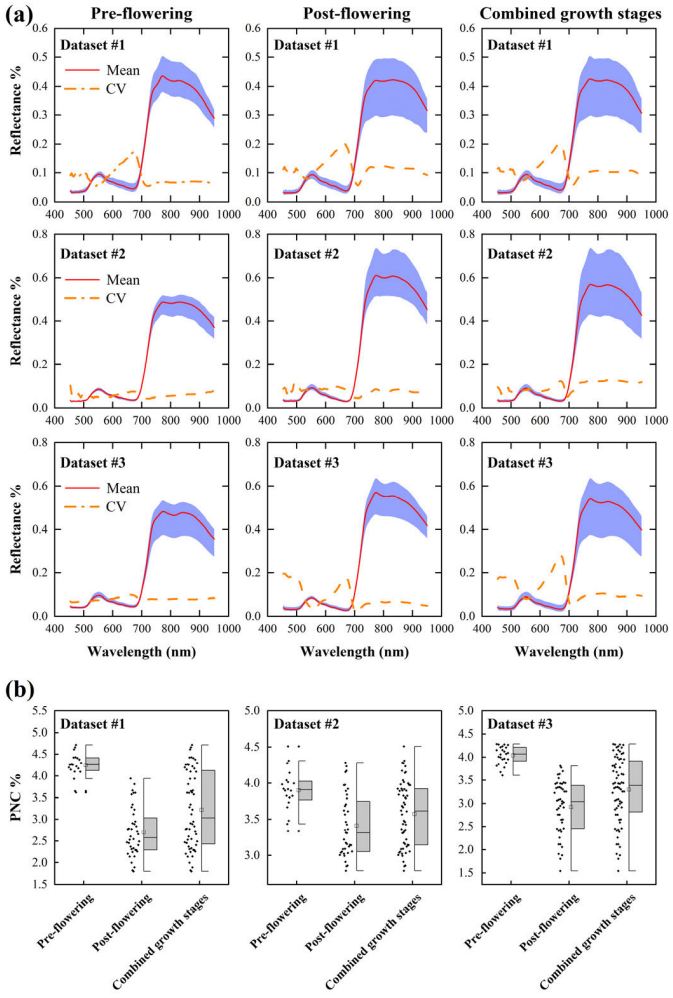



Fig. 4. (a) Variation of potato canopy reflectance and (b) PNC in different datasets and growth stages.

direct spectral band analysis, and its effectiveness with small datasets [8], [12]. In contrast, while nonlinear models or deep learning offer powerful modeling capabilities, they demand large datasets and remain a "closed box." Therefore, we evaluated the transferability of PLSR in estimating potato PNC.

PLSR establishes a regression model by mapping the original high-dimensional feature variables $X$ (hyperspectral reflectance) and response variables $Y$ (PNC) to a set of latent variables (LVs). Its basic form is as follows:

$$X = TP^T + E \tag{1}$$

$$Y = Tq + f \tag{2}$$

where $X$ is the original feature matrix (hyperspectral reflectance), $Y$ is the response variable (PNC), $T$ is the score matrix, $P$ is the load matrix describing the relationship between features and LVs, $q$ is the regression coefficient vector, and $E$ and $f$ are residual terms representing the part not explained by the model. The final regression model can be expressed as follows:

$$Y = XW\left(P^T W\right)^{-1} q \tag{3}$$

where $W$ is the weight matrix used to extract LVs from the original variables.

In hyperspectral remote sensing, raw spectral features are often high-dimensional and highly correlated, and direct modeling may lead to "dimension disaster" and multicollinearity issues [28]. PLSR extracts LVs to map high-dimensional raw data to a set of low-dimensional, uncorrelated new feature spaces, effectively compressing data dimensions while retaining the most relevant information for response variables (PNC). PLSR extracts LVs through an iterative algorithm,

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TABLE II
DIFFERENT MODELS USED IN THIS STUDY

| Model | Explanations | Parameter settings/optimization ranges |
|---|---|---|
| PLSR | Partial least squares regression as the baseline models for model transferability comparison | *LV* (the number of latent variables) optimization range: 1-20 |
| PLSR-TCA | PLSR coupled with the transfer component analysis (TCA), a featured-based transfer learning method | *LV* optimization range: 1-10<br>*m* (dimension after dimensionality reduction): 10<br>*Kernel function*: radial basis function (RBF)<br>*g* (the kernel parameter): 0.5 |
| PLSR-random | PLSR-random models by integrating 5%-50% target dataset samples into the source dataset | *LV* optimization range: 1-20 |
| PLSR-active | PLSR-active models by iteratively integrating the 1-10 target dataset samples with the largest residuals into the source dataset | *LV* optimization range: 1-20 |

with each LV constructed in the direction that maximizes the covariance between feature $X$ and response variable $Y$. The extraction process of the $i$th LV can be described as follows:

$$\mathrm{LV}_i = XW_i \tag{4}$$

$$W_i = \arg\max\left(\mathrm{Cov}\left(Xw, y\right)\right). \tag{5}$$

The variable importance in projection (VIP) value is an essential indicator of the contribution of each original variable to the estimation ability of the PLSR model [29]. By comprehensively considering the projection weights of each variable in all LVs and the explanatory power of these LVs on the response variable, the VIP value can reflect the global importance of variables in the modeling process. The VIP value is used to evaluate the contribution of each raw variable to the model-building process. The calculation formula is as follows:

$$\mathrm{VIP}_i = \sqrt{p \cdot \frac{\sum_{i=1}^{a}\left(SSY_i \cdot W_{ij}^2\right)}{\sum_{i=1}^{a} SSY_i}} \tag{6}$$

where $p$ represents the total number of features, $a$ represents the number of LVs extracted, $SSY_i$ represents the variance of the response variable explained by the $i$th LV pair, and $W_{ij}$ represents the weight of the $j$th variable in the $i$th LV. The optimal number of LV for the PLSR model was determined using Bayesian optimization, with a search range set from 1 to 20 (Table II). The PLSR model was run using the *PLSRegression* library in *Python 3.9*.

To further demonstrate the advantages of PLSR in PNC estimation, we compared it with both linear and nonlinear machine learning models. The linear models included regularized ridge regression (RR). The nonlinear models included distance-based $k$-nearest neighbors (KNNs), kernel-based SVR, ensemble learning-based random forest (RF), probabilistic estimation-based Gaussian process regression (GPR), and deep learning-based artificial neural network (ANN). Hyperparameter optimization, input parameters, and model transferability were consistent with the PLSR model.

### F. Transfer Learning Method

*1) Transfer Component Analysis:* TCA is an unsupervised domain adaptation algorithm [16]. It minimizes the maximum mean discrepancy (MMD) between the distributions of the transformed source and target domains. We applied TCA on both source and target domains using radial basis function (RBF) kernels with a kernel parameter 0.5. The features $X_{\mathrm{source}}$ and $X_{\mathrm{target}}$ were transformed into a new feature space $X'_{\mathrm{source}}$ and $X'_{\mathrm{target}}$ with ten dimensions. PLSR was then established on $(X'_{\mathrm{source}}, Y_{\mathrm{source}})$ and tested on $(X'_{\mathrm{target}}, Y_{\mathrm{target}})$ [7]. The application of TCA on cross-year datasets is shown in Fig. 5(a). The model parameter settings are shown in Table II. The TCA was performed in *Python 3.9*.

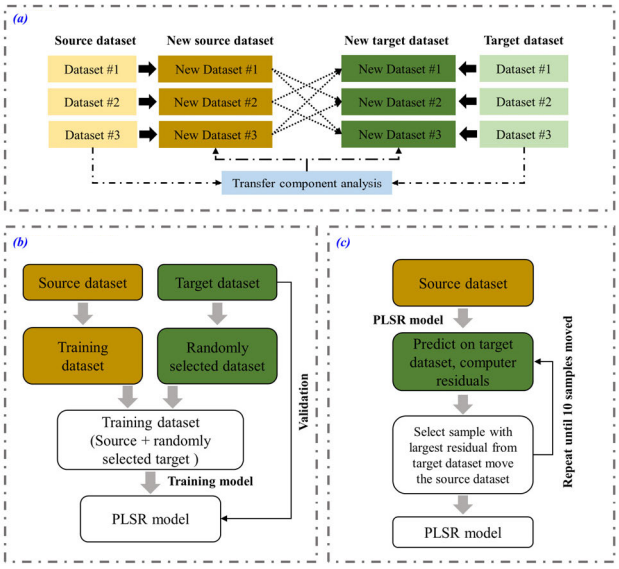



Fig. 5. (a) Detailed architecture of the TCAs, (b) random sample update method, and (c) active learning sample update method.

*2) Random Sample Update Method:* Random sample update method that transferred new samples from the target dataset ($X_{\mathrm{target}}$) to the source dataset ($X_{\mathrm{source}}$). Canopy reflectance of the new datasets is defined as follows:

$$X'_{\mathrm{source}} = X_{\mathrm{source}} \bigcup X_j \tag{7}$$

$$X'_{\mathrm{target}} = X_{\mathrm{target}} - X_{\mathrm{target}} \bigcap X_j \tag{8}$$

where $X'_{\mathrm{source}}$ and $X'_{\mathrm{target}}$ are the new source and target datasets after the model updating, respectively, $X_j$ is the transferred sample. Due to the limited amount of data between different years in cross-year estimation tasks, we randomly select 5%–50% of the samples from the target dataset and add them to the source dataset. Then, we recalibrate the PLSR model using the updated dataset ($X'_{\mathrm{source}}$). Finally, we apply the

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recalibrated model to the new target dataset ($X'_{\text{target}}$) to evaluate the model's transferability [Fig. 5(b)]. The model parameter settings are shown in Table II.

*3) Active Learning Sample Update Method:* Active learning can effectively address the uncertainty of random sample selection. This study combined active learning with transfer strategies to enhance the model's transferability across various conditions. The active learning sample update method, based on the maximum residual, gradually transfers high-residual samples from the target dataset ($X_{\text{target}}$) to the source dataset ($X_{\text{source}}$). Canopy reflectance of the updated dataset is defined as follows:

$$X'_{\text{source}} = X_{\text{source}} \bigcup_{j \in R_{\max}} X_j \tag{9}$$

$$X'_{\text{target}} = X_{\text{target}} - X_{\text{target}} \bigcap_{j \in R_{\max}} X_j \tag{10}$$

where $X'_{\text{source}}$ and $X'_{\text{target}}$ are the updated source and target datasets, respectively, and $R_{\max}$ denotes the index set of target samples with the most significant model residuals. In each iteration, we evaluate the estimation residuals of the initial PLSR model on the target dataset, rank all samples by residual magnitude, and select the sample with the highest residuals as the most informative sample. These selected samples $X_j$ are then transferred to the source dataset $X_{\text{source}}$ to form an updated training set. The PLSR model is subsequently retrained using $X'_{\text{source}}$, and the process is repeated for a predefined number of iterations or until model performance converges [Fig. 5(c)]. The residual-driven active learning strategy ensures that the most uncertain target samples are progressively included in the training process, thereby enhancing the model's ability to generalize across domain shifts. Finally, the recalibrated model is evaluated on the remaining target dataset ($X'_{\text{target}}$) to assess cross-year transferability. The model parameter settings are shown in Table II.

### G. PNC Estimation Model Input Parameters

The baseline PLSR, PLSR-random, and PLSR-active models used full-spectrum (450–950 nm) extracted in Section II-B as input parameters. Full spectral information was directly employed to take advantage of PLSR's capability in handling high-dimensional and collinear data. Spectral vegetation indices were not used in this study, as full-spectrum reflectance provides more comprehensive information and avoids potential information loss and subjectivity in index selection, which is particularly important for cross-year transfer learning. For the PLSR-TCA model, the input variables consisted of the transfer components obtained after TCA, as described in Section II-F1. These components represent the most transferable spectral features between source and target domains.

### H. Model Performance Evaluation

To more easily explain the accuracy of the estimation model, we used three evaluation parameters: the coefficient of determination ($R^2$), root mean square error (RMSE), and residual prediction deviation (RPD), to measure the model's performance. The higher the $R^2$, the smaller the RMSE, and the larger the RPD, indicating that the model has better estimation ability. The calculation formulas for the $R^2$, RMSE, and RPD are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_a - y_b)^2}{\sum_{i=1}^{n} (y_a - y_c)^2} \tag{11}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (y_a - y_b)^2}{N}} \tag{12}$$

$$\text{RPD} = \frac{\text{SD}}{\text{RMSE}} \tag{13}$$

where $y_a$ represents the measured value of PNC, $y_b$ represents the estimated value of PNC, $y_c$ represents the average of the measured samples, $N$ represents the number of samples, and SD represents the standard deviation of the measured value of PNC.

## III. Results

### A. Optimized Training Datasets With Random and Active Update Methods

To optimize the PLSR model training dataset, two sample updating methods, random and active methods, were applied to assess their effects on cross-dataset estimation performance (Fig. 6). Under the random sampling strategy, the model's RMSE generally declined as the proportion of transferred samples increased and tended to stabilize once the proportion exceeded approximately 30% [Fig. 6(a)]. For the combined growth stages, when the PLSR model was updated from Dataset #1 to Dataset #2 with 5%–50% transferred samples, the corresponding RMSE values were 0.59%, 0.56%, 0.50%, 0.45%, 0.40%, 0.34%, 0.30%, 0.31%, 0.32%, and 0.30%, respectively. When approximately 35% of the samples were transferred, certain transfer tasks achieved optimal accuracy, with RMSE values of 0.30% and 0.50% for transfers from Dataset #1 to Datasets #2 and #3, respectively. However, when the transfer proportion exceeded 35%, some transfer tasks exhibited an error rebound—for example, transferring 35%–50% of samples from Dataset #1 to Dataset #2 increased RMSE to 0.32%. This phenomenon likely results from amplified distribution discrepancies between the source and target domains, as well as instability in model parameters caused by the excessive inclusion of target-domain samples.

Under the active learning updating strategy, high-residual samples identified by the PLSR model were progressively added to the training dataset. The RMSE demonstrated an initial fluctuation followed by a steady decline and stabilization [Fig. 6(b)]. At the early transfer stage, uncertainty associated with selected samples caused noticeable error oscillations. When the number of transferred samples reached approximately 5–6, RMSE reached its minimum across most dataset combinations; subsequent additions induced slight variations but overall remained stable. Specifically, during the preflowering stage of potato growth, transferring five high-residual samples achieved the lowest RMSE (0.09%–0.28%). In contrast, during the postflowering and combined stages, transferring six high-residual samples yielded the lowest RMSE (0.24%–0.55%). These results indicate that active learning can substantially reduce the number of required

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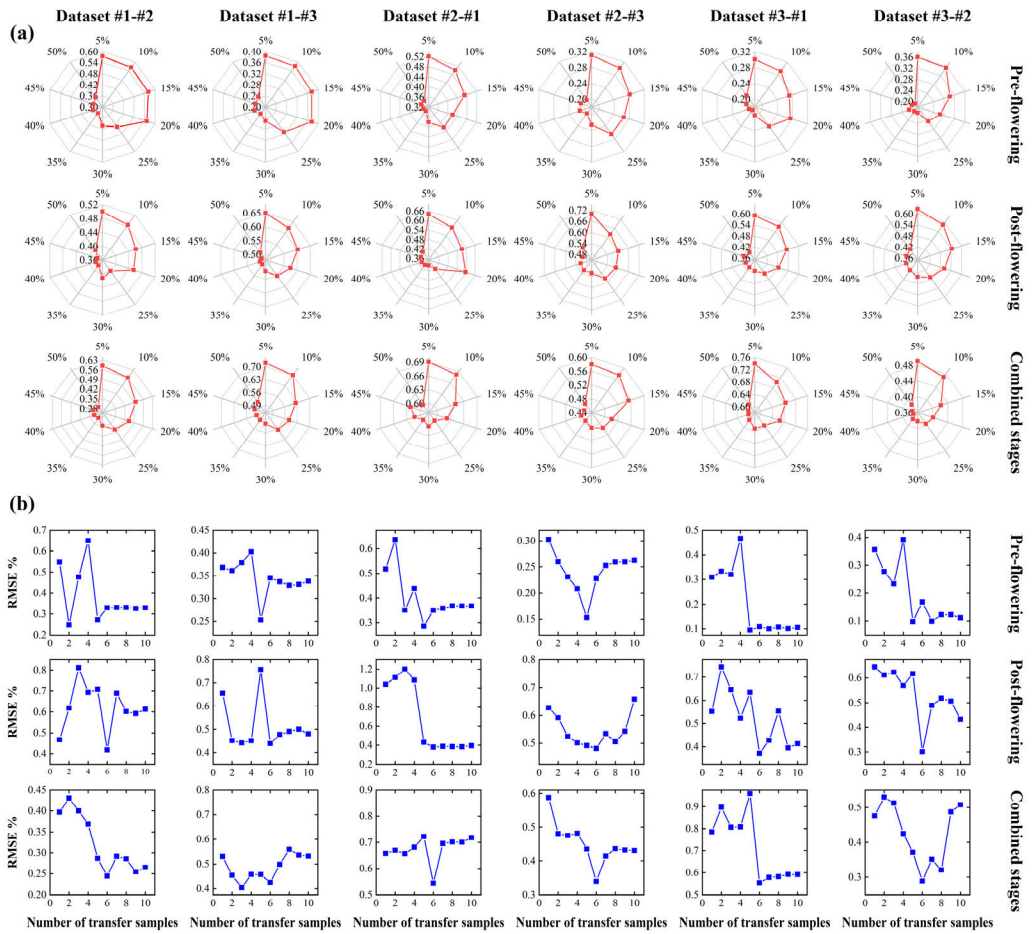



Fig. 6. (a) Random sample update based on the PLSR model on different datasets. The model is optimized according to the different proportions of the target dataset, and the proportion of the transformed samples is displayed. (b) Active learning sample update based on the PLSR model on different datasets. The model is optimized based on the magnitude of the residual values, and the sample size of each transfer is displayed. "Datasets #1 and #2" indicate using Dataset #1 as the training set and Dataset #2 as the validation set.

transfer samples while maintaining or even improving model estimation accuracy and stability.

### *B. Transfer Learning Model for Assessing PNC Over Different Datasets*

To assess the cross-dataset performance of different transfer approaches, three transfer learning methods were systematically compared with the conventional PLSR model (Table III). The results indicated that TCA provided only marginal improvements in PLSR transferability. In most cases, the RMSE of the PLSR-TCA model was comparable to that of the conventional PLSR model and exhibited superior performance only when transferring from Dataset #1 to Datasets #2 and #3, where the RMSE decreased from 0.41% to 1.13% (PLSR) to 0.31% to 0.51% (PLSR-TCA). In contrast, sample-updating-based transfer methods consistently outperformed PLSR-TCA, with the PLSR-active model demonstrating superior performance. During the preflowering stage of potato, RMSE was reduced by 11.5%–39.39% using PLSR-random and by 3.8%–72.7% using PLSR-active, relative to the conventional PLSR model. Owing to pronounced differences in sample distributions, the performance gain of PLSR-active was limited when transferring from Dataset #1 to Dataset #3, with an RMSE reduction of only 3.8%. During the postflowering stage, both PLSR-random and PLSR-active exhibited substantial error decreases across most dataset combinations. The largest improvements were observed for transfers from Dataset #2 to Dataset #1 and from Dataset #3 to Dataset #2, with RMSE reductions of 42.6% and 32.3% for PLSR-random and 45.6% and 51.6% for PLSR-active, respectively. During the combined growth stage, both methods achieved significant reductions in RMSE across all transfer tasks, with PLSR-active showing the greatest improvements (31.3%–62.8%), followed by PLSR-random (23.8%–55.8%). Overall, among the three evaluated transfer learning methods, the PLSR-active model demonstrated the most robust estimation performance, substantially enhancing model transferability and generalization capability across heterogeneous datasets.

### *C. Effect of Transfer Learning on PLSR Model Parameters*

To further investigate the influence of different transfer learning approaches on model internal structures, the Dataset #2 to Dataset #1 transfer task was taken as a representative example, and the distributions of samples generated by the conventional PLSR, PLSR-TCA, PLSR-random, and PLSR-active models were examined in the space defined by the first

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TABLE III
PNC IS EVALUATED ON DIFFERENT DATASETS USING THE PLSR, PLSR-TCA, PLSR-RANDOM, AND PLSR-ACTIVE MODELS

| Training | Validation | Models | Pre-flowering stage | | | Post-flowering stage | | | Combined growth stages | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $R^2$ | RMSE% | RPD | $R^2$ | RMSE% | RPD | $R^2$ | RMSE% | RPD |
| Dataset #1 | Dataset #2 | PLSR | 0.27 | 0.42 | 0.69 | 0.44 | 0.41 | 1.04 | 0.28 | 0.58 | 0.77 |
| | | PLSR-TCA | 0.47 | 0.40 | 0.72 | 0.45 | 0.33 | 1.30 | 0.49 | 0.33 | 1.36 |
| | | PLSR-random | 0.57 | 0.34 | 0.67 | 0.63 | 0.38 | 0.88 | 0.60 | 0.30 | 1.42 |
| | | PLSR-active | 0.62 | 0.27 | 0.78 | 0.65 | 0.40 | 0.88 | 0.68 | 0.24 | 1.77 |
| Dataset #1 | Dataset #3 | PLSR | 0.22 | 0.26 | 0.75 | 0.26 | 0.71 | 0.81 | 0.20 | 1.13 | 0.63 |
| | | PLSR-TCA | 0.40 | 0.18 | 1.09 | 0.44 | 0.63 | 0.91 | 0.37 | 0.59 | 1.21 |
| | | PLSR-random | 0.48 | 0.23 | 0.50 | 0.63 | 0.50 | 0.84 | 0.52 | 0.50 | 1.30 |
| | | PLSR-active | 0.50 | 0.25 | 0.53 | 0.75 | 0.43 | 1.03 | 0.68 | 0.42 | 1.42 |
| Dataset #2 | Dataset #1 | PLSR | 0.30 | 0.42 | 0.68 | 0.60 | 0.68 | 0.83 | 0.29 | 0.98 | 0.89 |
| | | PLSR-TCA | 0.37 | 0.40 | 0.71 | 0.63 | 0.39 | 1.45 | 0.47 | 0.78 | 1.12 |
| | | PLSR-random | 0.43 | 0.34 | 0.75 | 0.63 | 0.39 | 1.43 | 0.58 | 0.60 | 1.23 |
| | | PLSR-active | 0.59 | 0.28 | 0.87 | 0.64 | 0.37 | 1.46 | 0.64 | 0.54 | 1.54 |
| Dataset #2 | Dataset #3 | PLSR | 0.13 | 0.23 | 0.85 | 0.18 | 0.62 | 0.93 | 0.29 | 0.70 | 1.02 |
| | | PLSR-TCA | 0.26 | 0.22 | 0.86 | 0.41 | 0.60 | 0.96 | 0.68 | 0.63 | 1.13 |
| | | PLSR-random | 0.40 | 0.20 | 1.10 | 0.53 | 0.52 | 0.95 | 0.70 | 0.47 | 1.35 |
| | | PLSR-active | 0.54 | 0.15 | 1.22 | 0.63 | 0.48 | 0.99 | 0.73 | 0.33 | 1.81 |
| Dataset #3 | Dataset #1 | PLSR | 0.28 | 0.29 | 0.98 | 0.22 | 0.52 | 1.08 | 0.18 | 0.80 | 1.10 |
| | | PLSR-TCA | 0.41 | 0.28 | 1.01 | 0.34 | 0.51 | 1.11 | 0.34 | 0.76 | 1.15 |
| | | PLSR-random | 0.47 | 0.19 | 1.12 | 0.45 | 0.40 | 1.24 | 0.45 | 0.61 | 1.38 |
| | | PLSR-active | 0.56 | 0.09 | 1.53 | 0.55 | 0.37 | 1.30 | 0.55 | 0.55 | 1.49 |
| Dataset #3 | Dataset #2 | PLSR | 0.17 | 0.33 | 0.87 | 0.33 | 0.62 | 0.69 | 0.21 | 0.55 | 0.82 |
| | | PLSR-TCA | 0.31 | 0.24 | 1.20 | 0.35 | 0.60 | 0.71 | 0.32 | 0.52 | 0.84 |
| | | PLSR-random | 0.42 | 0.20 | 1.25 | 0.40 | 0.42 | 1.12 | 0.43 | 0.38 | 1.25 |
| | | PLSR-active | 0.63 | 0.09 | 1.84 | 0.48 | 0.30 | 1.31 | 0.61 | 0.28 | 1.58 |

PLSR-TCA: Transfer component analysis-based PLSR model; PLSR-random: Random sample update-based PLSR model; PLSR-active: Active learning sample update-based PLSR model.

two LVs (LVs1 and LV2) [Fig. 7(a)]. In addition, differences in VIP across the full spectral range between the PLSR and PLSR-active models were systematically compared [Fig. 7(b)] by analyzing the corresponding VIP values. In the LV space, the conventional PLSR model exhibited a clear separation between source-domain and target-domain samples, particularly during the preflowering and postflowering stages, where samples formed distinct clusters. This pattern indicates the limited capacity of the conventional PLSR model to accommodate spectral discrepancies across datasets. The PLSR-TCA model partially improved this distribution by reducing intersample distances; however, pronounced class boundaries remained. The PLSR-random model produced a more dispersed pattern, with partially weakened class boundaries but still limited clustering compactness. In contrast, the PLSR-active model achieved the most effective LV alignment, with substantial overlap between source and target samples across all three growth stages, and more compact distributions. This demonstrates that the active learning-based sample updating strategy markedly enhanced the consistency of spectral representations across datasets.

Analysis of band contributions revealed substantial differences in model sensitivity to spectral information. In the conventional PLSR model, most VIP values were concentrated around the threshold (VIP = 1), and important bands were scattered with limited stability. Conversely, the PLSR-active model showed higher and more stable VIP responses across several key spectral regions, especially within the 520–570 nm and 680–730 nm ranges. These regions closely correspond to well-established nitrogen-sensitive spectral domains in plants, indicating that the important bands identified by the PLSR-active model are more physiologically meaningful and better aligned with the true spectral characteristics of crop nitrogen status.

### D. Upscaling UAV-Based PNC Mapping to the Sentinel-2A Imagery

Fig. 8(a) shows the temporal and spatial variations of the estimated PNC of potato in Dataset #1. The potato PNC exhibits a gradual downward trend, corresponding to changes in the growth stages. The PNC map is consistent with the changes in the measured PNC. The PNC distribution map can help farmers observe changes in PNC in the field and assist in applying $N$ fertilizer. To demonstrate the mapping capabilities of the PLSR-active model, it was executed on Sentinel-2A imagery covering the entire test area in Dataset #3. Fig. 8(b) shows the resulting PNC map. We trained the PLSR-active model using UAV data and then updated it with Sentinel-2A imagery data. The results show that the Sentinel-2A-based PNC distribution map can effectively reflect the spatial variability of PNC ($R^2$ = 0.67, RMSE = 0.58, and RPD = 1.22). Due to the "dilution effect," the PNC shows a gradual downward trend. This demonstrates that the PLSR-active model can be applied across sensor platforms and provides a promising method for mapping regional PNC.

## IV. DISCUSSION

Hyperspectral remote sensing technology, combined with machine learning algorithms, has been widely used for the dynamic monitoring of PNC [5], [30], [31], [32]. However, changes in growth conditions lead to poor performance of

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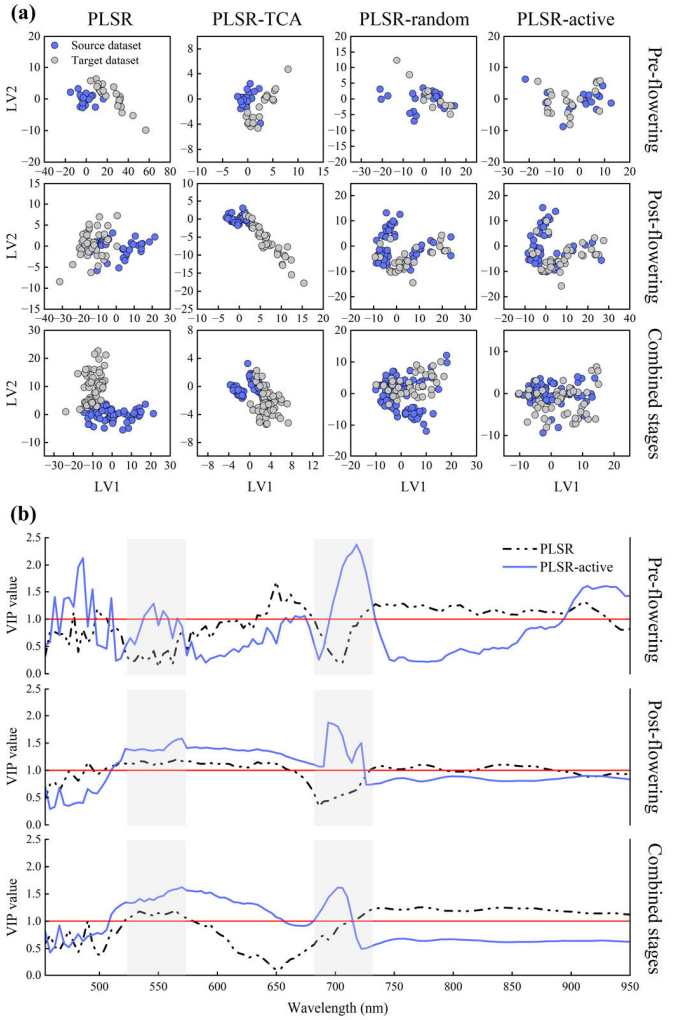



Fig. 7. (a) Data distribution of the first two PLSR LVs (LV1 and LV2) in PLSR, PLSR-TCA, PLSR-random, and PLSR-active. (b) Model VIP value difference between PLSR and PLSR-active models.

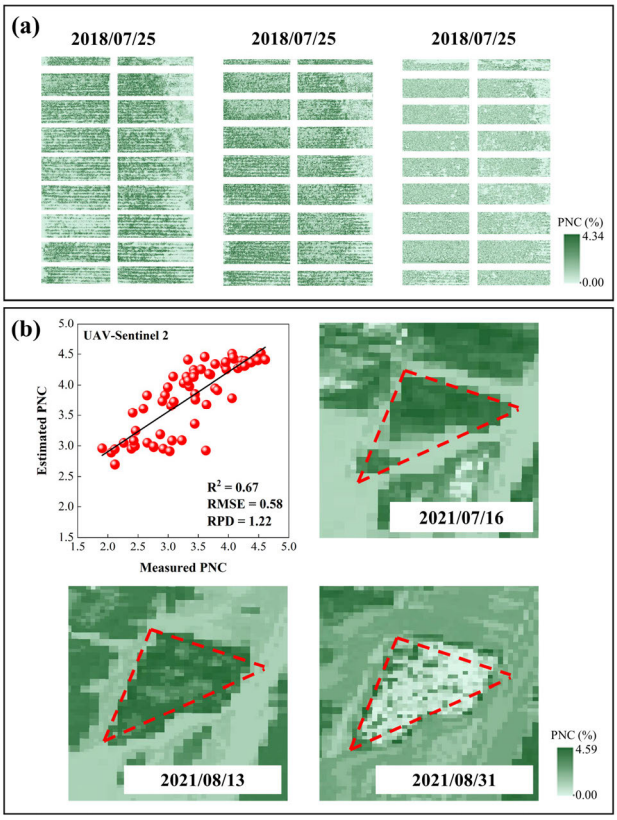



Fig. 8. (a) PLSR-active-based field distribution graphic showing potato PNC estimated values at different growth stages. (b) UAV image data are transferred to Sentinel-2A images using PLSR-active model. The model is trained using UAV image data from 2020 and validated using satellite image data from 2021.

hyperspectral models and thus have always limited the practical application [33], [34]. This study proposed an active learning-guided transfer learning strategy to enhance the transferability of hyperspectral models across years. The proposed method is compared with traditional transfer learning strategies through comparative analysis. The results demonstrate that the active learning-guided transfer learning strategy plays a crucial role in the remote estimation of crop PNC.

### A. *Unsupervised-Based Transfer Learning Models*

Traditional PLSR performed poorly at deriving the PNC in this study. The model has the lowest estimation accuracy across years (Table III). This may be due to differences in spectral distribution among years. Fig. 7 also demonstrates that the distributions of the first two LVs in the PLSR model differ significantly. PLSR models are typically constructed under the assumption that the distributions of the source and target datasets are the same. However, when the spectral characteristics of the target dataset change, the model may not be able to adapt to the new data patterns. This is why using an independent validation dataset from the different years to evaluate the model's performance is less effective than using the randomly split dataset or cross-validation. Another reason is the small amount of training data used to construct the PLSR model. For example, the model may be trained well on Dataset #1, but when Dataset #2 changes, the model may not be able to make reasonable estimation for Dataset #2 (Table III). Wan et al. [16] used TCAs to partially address this issue to reduce the spectral distribution differences between the source and target datasets, thereby improving the model's estimation accuracy across years and species. Similarly, the results of this study show that the PLSR-TCA can mitigate the impact of data distribution differences between years and effectively enhance the generalization ability of the potato PNC estimation model. However, TCA does not entirely eliminate the differences in the distributions of data in the source and target datasets. When the quality of the source dataset model performance significantly. As shown in Fig. 4, there are significant differences in the spectral information and measured PNC values between datasets from different years. Overall, there is still no best approach to effectively improve the model's generalization ability.

### B. *Sample-Based Transfer Learning Models*

Sample-based transfer learning strategies can effectively improve model estimation accuracy across years. Previous studies have mainly focused on methods that randomly select

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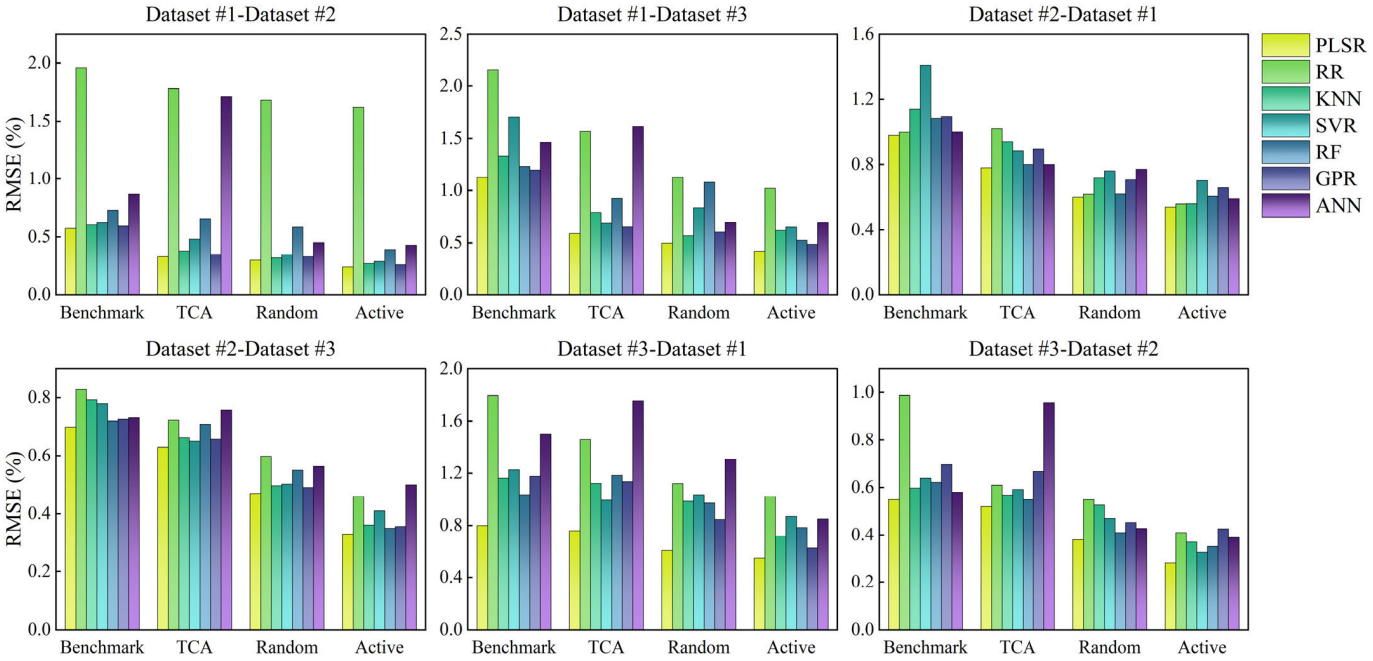



Fig. 9. Comparison of PLSR with different machine learning models during the potato combination growth stages. Benchmark: original baseline model, TCA: transfer component analysis combined with machine learning models, random: random sample update method combined with machine learning models, active: active sample update method combined with machine learning models, PLSR: partial least squares regression, RR: ridge regression, KNN: *k*-nearest neighbors, SVR: support vector regression, RF: random forest, GPR: Gaussian process regression, and ANN: artificial neural network.

a portion of the target dataset to update the source dataset. Once the target dataset is integrated into the source dataset, the model can learn the patterns of different distributions, thereby improving the model's generalization ability. For example, Wan et al. [16] used a random sample update method to update the source dataset with 7% of the target dataset samples, which was shown to effectively improve the generalization ability of their leaf nitrogen content estimation model on various datasets. However, the random sample update method also has its drawbacks. Crucial samples in the target dataset may be ignored or rarely selected, hindering the model's ability to capture essential changes in the target dataset. In addition, random processes may include noisy or unrepresentative samples, which can introduce bias or disrupt the learning process, thereby reducing the model's performance when applied to data in the target dataset [35].

In our study, this phenomenon was reflected by the observation that model accuracy no longer monotonically increased and even slightly decreased when the random sample proportion exceeded a certain threshold (Fig. 6). This indicates that simply increasing the quantity of transferred samples does not guarantee performance improvement and may introduce uncertainty when the sampling quality is not controlled. Compared with random updating, the active learning-based sample update strategy demonstrated superior efficiency and robustness. Active learning prioritizes samples with the largest estimation residuals, which are considered the most informative and uncertain samples. In this study, transferring only a small number of high-residual samples resulted in substantial improvements in model performance (Table III). This confirms that selecting informative samples is more effective than random sampling when adapting models across years. This is consistent with the findings of Verrelst et al. [22], who demonstrated that active learning can effectively improve the model's generalization ability, especially in dynamic environments where new data may not always conform to previous patterns. More importantly, in the active learning process, we observed that irregular fluctuations in RMSE mainly occurred when only a small number of transfer samples were incorporated (Fig. 6). This phenomenon is closely related to the mechanism of active learning. In the early stage of sample transfer, only a few high-residual samples were selected, and these samples often represented extreme, atypical, or highly heterogeneous conditions. The limited number of transferred samples was therefore insufficient to stabilize the model, resulting in nonsmooth and irregular variations in RMSE across different years. As the number of actively transferred samples gradually increased, the training dataset became more representative of the target domain, and the model performance became progressively more stable. When a sufficient number of informative samples had been incorporated, the RMSE curves tended to exhibit smoother trends and stabilized performance across different years. This indicates that active learning requires an adequate number of transferred samples to balance sample informativeness and distribution representativeness, thereby achieving robust cross-year generalization. Additionally, we applied different transfer learning methods to multiple machine learning models, and the results demonstrate that active transfer learning consistently achieves performance improvements across all models (Fig. 9). Among these, PLSR demonstrated the best performance across all models, proving its superiority as a baseline model for transfer learning.

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### C. Cross-Scale Transfer Modeling From UAV to Satellite

Satellite images have been widely used for large-scale PNC estimation. In recent decades, some studies have directly used satellite remote sensing data to estimate crop PNC [22], [23], [36]. However, limited studies have linked UAV images with satellites. Satellite imagery has a long revisit cycle and is significantly affected by weather conditions [37], [38]. In contrast, UAV imagery can effectively address these limitations. This study suggests that the PNC estimation model, trained on UAV image data, can be applied to satellite images (Fig. 8). This may be attributed to the potential of active learning for cross-platform applications. Furthermore, active learning to develop cross-platform applications can also aid in designing and developing small spectral sensors. Using spectral sensor technologies, farmers can be guided to apply $N$ at the correct rate and time [39].

While the active learning sample update method demonstrates improved estimation performance, several limitations persist in the current framework. The study area in this research was geographically limited, and future work should further investigate the applicability and scalability of the proposed model across broader agroecological regions. While this study provided preliminary evidence of the model's transferability from UAV-based hyperspectral imagery to Sentinel-2A satellite data, it remains necessary to develop a more systematic framework for multi-source remote sensing data fusion to support regional and global-scale applications [20]. Incorporating datasets from multiple locations, climate zones, and soil types for cross-validation will help enhance model robustness under diverse ecological and management conditions. Furthermore, building an integrated sensing system that leverages data from UAV, satellites, and ground-based sensors offers a promising pathway for large-scale crop nitrogen monitoring [40], [41]. Such an approach also aligns with the ongoing evolution of remote sensing toward integrated "ground–air–space" observation systems and supports the broader agenda of agricultural digitalization at the global scale.

## V. Conclusion

This study demonstrated the effectiveness of an active learning-guided transfer learning strategy in enhancing the cross-year transferability of models estimating PNC. Our comprehensive evaluation revealed that while all three transfer strategies (PLSR-TCA, PLSR-random, and PLSR-active) improved the transferability of the baseline PLSR model, PLSR-active consistently achieved superior performance with the lowest RMSE across years. Compared with the PLSR-random method, which requires more transfer samples, the PLSR-active method achieves optimal performance with fewer than ten samples, demonstrating significant efficiency. Feature distribution analysis confirmed that PLSR actively minimized the gap between source and target datasets. In contrast, band contribution analysis showed enhanced sensitivity in the critical 520–570 nm and 680–730 nm regions, better aligning with nitrogen's spectral absorption characteristics. The active learning-guided transfer learning strategy proposed in this study not only significantly enhanced model generalization across multiyear datasets but also exhibited strong potential for cross-platform transfer between UAV-based and satellite imagery. Future research may further extend this approach to diverse regions and crop types by integrating multi-source remote sensing data to develop a generalized framework for PNC estimation. The broader adoption of such methods will support large-scale, high-precision monitoring of crop nutritional status and facilitate the deployment of precision agriculture technologies across varied geographic and ecological contexts.

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